

‘Catalytic’ doses of fructose may benefit glycaemic control without harming cardiometabolic risk factors: a small meta-analysis of randomised controlled feeding trials

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Contrary to concerns that fructose may have adverse metabolic effects, there is evidence that small, ‘catalytic’ doses (≤ 10 g/meal) of fructose decrease the glycaemic response to high-glycaemic index meals in human subjects. To assess the longer-term effects of ‘catalytic’ doses of fructose, we undertook a meta-analysis of controlled feeding trials. We searched MEDLINE, EMBASE, CINAHL and the Cochrane Library. Analyses included all controlled feeding trials ≥ 7 d featuring ‘catalytic’ fructose doses (≤ 36 g/d) in isoenergetic exchange for other carbohydrates. Data were pooled by the generic inverse variance method using random-effects models and expressed as mean differences (MD) with 95 % CI. Heterogeneity was assessed by the Q statistic and quantified by I². The Heyland Methodological Quality Score assessed study quality. A total of six feeding trials (n 118) met the eligibility criteria. ‘Catalytic’ doses of fructose significantly reduced HbA_{1c} (MD – 0.40, 95 % CI – 0.72, – 0.08) and fasting glucose (MD – 0.25, 95 % CI – 0.44, – 0.07). This benefit was seen in the absence of adverse effects on fasting insulin, body weight, TAG or uric acid. Subgroup and sensitivity analyses showed evidence of effect modification under certain conditions. The small number of trials and their relatively short duration limit the strength of the conclusions. In conclusion, this small meta-analysis shows that ‘catalytic’ fructose doses (≤ 36 g/d) may improve glycaemic control without adverse effects on body weight, TAG, insulin and uric acid. There is a need for larger, longer (≥ 6 months) trials using ‘catalytic’ fructose to confirm these results.